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Participatory Design for Small-Scale PV Integration in Heritage Districts: The Case of Öjeby Church Town, Piteå, Sweden

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Abstract

The integration of small-scale photovoltaic (PV) systems in heritage districts poses a significant challenge: balancing sustainability and energy transition goals while preserving cultural and historical values. This study addresses the problem that existing planning and regulatory frameworks often exclude residents' perspectives, leading to solutions that are technically feasible but socially contested. The objective is to explore how residents, as potential prosumers, can be effectively integrated into decision-making processes for PV adoption in heritage districts. Focusing on Öjeby Church Town in Piteå, northern Sweden, we employed transdisciplinary participatory design methods, including stakeholder workshops, interviews, council meetings, and a tailored resident design workshop to capture both explicit and tacit knowledge. These design methods were combined with spatial analysis and visual assessment. The findings reveal that residents favour PV solutions that minimise visual impact on heritage buildings, preferring installations in green and grey infrastructure over direct building integration. The process also enhanced awareness, legitimacy, and agency among participants, while exposing regulatory gaps, ownership complexities, and aesthetic tensions. The study contributes a replicable participatory framework that integrates community values with technical and heritage expertise, offering a pathway toward value-sensitive energy transitions in protected cultural environments.

Keywords: participatory design; urban energy transition; cultural heritage districts; photovoltaics; prosumers

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1. Introduction

Like the rest of the European Union (EU), Sweden has set ambitious goals for renewable energy [1]. The EU has established a binding renewable energy target of at least 42.5% by 2030, intending to reach 45%. This initiative is part of the broader objective to achieve climate neutrality by 2050 [2]. In 2017, Sweden adopted a climate policy framework with a long-term goal of achieving net-zero emissions of greenhouse gases by 2045 at the latest, followed by achieving negative emissions [3].

The Swedish energy system has historically been characterised by large-scale, centralised production, primarily relying on hydropower and nuclear power. This centralised structure facilitates the flow of electricity from a few large producers to consumers [4]. However, governmental and regulatory bodies have begun to recognise the importance of decentralised energy systems by promoting small-scale, distributed projects through various incentives and supportive regulations [5].

In recent years, the production of energy from small-scale photovoltaic (PV) systems has significantly increased in Sweden. As of 2023, over 100,000 PV systems were connected to the national grid, with a combined capacity of approximately 1600 megawatts, representing a nearly 70% increase compared to 2022 [6]. Currently, approximately 250,000 PV systems are installed nationwide.

It is recognised that heritage districts, like any other urban areas, must respond to contemporary societal demands by meeting energy needs, adapting to current expectations, and pursuing greater sustainability [7,8]. While some studies advocate minimising the impact on historical buildings by installing photovoltaic (PV) systems on adjacent non-historical structures to satisfy district-level energy requirements [9], this approach may not always be viable. In many cases, particularly during renovation or retrofit projects, there is potential to improve a building's energy performance directly [10,11]. In such scenarios, building-integrated photovoltaic (BIPV) systems present a promising solution, such as transparent photovoltaic films, which can be aesthetically and functionally integrated into the fabric of existing structures without compromising their heritage value [12–16].

However, building-applied photovoltaics (BAPV) remain the most prevalent form of PV integration [17], highlighting the ongoing tension between technological advancement and heritage preservation. PV may significantly alter the visual character of historic buildings; existing regulations can limit or prohibit integration, creating a conflict between environmental sustainability and heritage preservation [18].

This tension highlights the importance of carefully considering the heritage values associated with historic buildings when integrating modern technologies, such as solar energy supply [19]. These values can be mitigated through historical and visual assessment approaches [20].

Previous studies generally fall into three main categories: technical compatibility, policy adaptation, and social acceptance. Most research on the integration of photovoltaic (PV) systems in heritage districts has primarily focused on central and southern European countries. Research within the technical compatibility category [21,22] often focuses on how PV systems can be physically integrated into buildings and outdoor environments. This focus may overlook broader community perspectives and the social dynamics surrounding heritage conservation.

Several studies emphasise the need for comparative research across diverse cultural settings, as well as interdisciplinary collaboration among technical experts, social scientists, and heritage professionals, to support more inclusive and context-sensitive approaches to PV integration [19,23–26]. A common theme across the studies is the need for clear policies, stakeholder engagement, and tailored solutions to address the unique challenges of integrating PV in different contexts. The studies also reveal significant regional differences in legislative and regulatory frameworks, which impact the effectiveness of solar energy adoption.

Additionally, a lack of knowledge about PV technology is not a significant barrier to adoption. The acceptance of PV systems is driven mainly by their ability to enhance the aesthetics of historical buildings and landscapes [23].

Research on photovoltaic (PV) integration in historical buildings in Sweden considers both reversibility and visual sensitivity [27]. Visibility approaches aim to ensure that PV systems are integrated without damaging or altering the appearance of protected features. PV installations should also be reversible, especially on listed buildings or those in culturally and architecturally sensitive areas [28]. Additionally, some municipalities in Sweden have developed clear guidelines for PV in heritage zones, especially in larger cities such as Stockholm and Malmö. These regulations have opportunities to strike a balance between aesthetic integrity and climate targets, providing criteria for placement,

colour matching, and impact assessment [29]. Still, smaller local authorities often lack these criteria.

Some studies have been done in the Nordic countries on social acceptance and stakeholders' perspectives. Factors such as housing density, homeownership rates, per-capita income, and neighbourhood characteristics significantly influence the adoption of photovoltaic (PV) systems [30]. Interestingly, the ecological attitudes of households have a minimal impact on their investment decisions. In Finland, households that adopt solar photovoltaic (PV) systems are primarily motivated by non-economic factors. Even if the financial returns may be limited, adopters derive satisfaction from generating clean energy and contributing to environmental sustainability [31].

Adopting ethical views on conservation and a community-engaged philosophy may be crucial for carefully integrating contemporary PV technology into traditional, older heritage districts. Heritage districts embody a community's identity and reflect the values of various stakeholders [32]. Muñoz Viñas emphasises the importance of subjective decisions and values in conservation, arguing that conservation depends on personal preferences, which are influenced by the prevailing tastes of a specific time or individual.

According to de la Peña et al. [33], participatory design techniques can empower communities through collective creativity and innovation. This approach values local knowledge, cultural context, and lived experiences as communities are not merely consulted but serve as co-creators of their environments. Participatory design practices also acknowledge and incorporate various values, recognising that stakeholders bring diverse perspectives and priorities to the design process [34]. This inclusivity enhances the relevance and responsiveness of architectural outcomes.

Arnstein [35] argues that residents are the true experts of their neighbourhoods, and their involvement is not merely beneficial but often necessary. Engaging end-users can help identify balanced solutions that respect traditional heritage values while enabling the adoption of PV technologies. Participatory design supports this objective by providing accessible tools and strategies, such as mock-ups, games, temporary interventions, and idea-sharing activities, that facilitate meaningful contributions from individuals regardless of their technical expertise or language proficiency [36,37].

Moreover, the cultural significance of historic buildings extends beyond experts and authorities to include residents. Suppose decision-making is limited to those with technical knowledge or professional authority. In that case, other stakeholders' risks are being excluded from discussions that directly impact their experience and connection to the built environment, and restricting participation to experts alone can undermine the democratic and pluralistic values essential to heritage conservation [38].

Accordingly, participatory design should be employed not only as a methodological tool but as an ethical imperative to ensure that PV integration in heritage contexts reflects the values, concerns, and aspirations of all affected communities.

In addition to consulting various professionals, it is essential to engage residents who aspire to become micro-energy producers. Their inclusion in the decision-making process not only fosters a sense of ownership but also enhances the relevance and social acceptance of design interventions.

The research problem of this study is that not all stakeholder perspectives are considered in the integration of small-scale PV systems in heritage districts, leading to outcomes that may not fully meet the social or technical requirements of these areas. Therefore, the study adopts a transdisciplinary approach, which is necessary to address the challenges of integrating PV systems in heritage districts.

Transdisciplinary approaches go beyond the academic realm by incorporating non-academic stakeholders, such as community members, policymakers, and a variety of

experts, in both the framing of problems and the creation of knowledge. This approach emphasises co-creation, collaboration, and a focus on real-world impact [39,40].

A transdisciplinary design process may require both resident design and design proposals, as well as design legibility. Resident design, which refers to the notion that local inhabitants play a primary role in articulating their vision for the future of their community [41]. A design proposal emerges when the vision articulated by residents is translated into more concrete forms through collaboration with professional actors [42]. Design legibility refers to the clarity, transparency, and comprehensibility of the proposed design, both for the community and for regulatory bodies [43].

One promising arena for developing a transdisciplinary participatory design process is the urban living lab (ULL) environment. ULLs offer experimental, real-life settings where researchers, policymakers, and community members collaborate to co-create, test, and evaluate innovative solutions for sustainable urban development. They emphasise iterative learning, cross-disciplinary collaboration, and user-centred experimentation, making them particularly suitable for addressing complex challenges such as the integration of renewable energy technologies in heritage-sensitive districts [44,45].

We suggest that actively engaging the local community, including residents and potential prosumers, is essential for a successful urban energy transition. For this reason, our objectives are to adopt participatory design methods to develop a transdisciplinary participatory design process for heritage districts. Our research question is: How can the perspectives of residents as prosumers be effectively integrated into decision-making processes for small-scale photovoltaic (PV) integration in heritage districts?

The research employs an integrated, participatory framework to investigate how energy transition may be reconciled with heritage preservation. The conceptualisation and operationalisation of this research follow a structured, multidimensional approach grounded in participatory design, cultural analysis, spatial planning, and technological feasibility.

The framing of the research problem involves determining the key concepts to explore and providing groundwork for integrating PV systems into heritage districts. The key concepts include perceptions of PV integration, which will be analysed through a tailored participatory approach. Ownership and governance structures, architectural typology, cultural values and traditions, and technological feasibility in a climatic context, as analysed through the case study.

2. Description of the Case Study

This case study has been undertaken to understand and explore the preconditions for PV integration within the context of heritage-sensitive areas. It serves as a lens through which multiple interacting systems—heritage, governance, ownership, architecture, and energy technology—can be studied.

2.1. Historical Context and Cultural Significance of Church Towns

Heritage theory underscores that places derive meaning from accumulated social, economic, and religious practices; interventions that ignore these dimensions risk cultural dissonance. Integrations are therefore more likely to succeed when aligned with the symbolic values and collective memory embedded in such spaces [46].

Church towns, as unique cultural landscapes in northern Scandinavia, illustrate the interplay between heritage, identity, and contemporary planning. Understanding Öjeby Church Town's historical layers and traditions is crucial for designing interventions that respect local identity and sustain living heritage.

Originating in the 17th century, these settlements were created to accommodate churchgoers travelling from remote villages during an era of mandatory church attendance. Journeys were often long and arduous, necessitating overnight stays [47]. Church Towns exemplify early forms of regional urban planning and social organisation, granting them substantial architectural and cultural-historical significance [48]. Today, their enduring value is recognised through formal protections as cultural heritage, national interest sites, and, in some cases, World Heritage Sites [49].

Historically, church towns served functions that extended beyond religion. They operated as hubs for trade and social interaction, reflecting their role as both sacred and communal spaces [50]. This multifunctional character persists today through temporary uses during religious holidays, cultural festivals, and markets. Many towns have preserved the tradition of providing accommodation for visitors during such events, reinforcing their relevance as living heritage [51].

2.2. Governance and Ownership Relations

To explore a sustainable path forward, it is crucial to examine the ownership relationships and governance structure of multiple co-owners with differing rights and responsibilities [52]. It emphasises the importance of trust, participation, and self-governance in promoting collective action and sustainable resource stewardship [53].

The church cottages are privately owned (either by purchase or inheritance) and used for short-term stays, often multiple times a year, where owners and relatives return for various gatherings [54].

However, there are complex ownership relations regarding the church cottages, which may affect the PV integration. The church cottages in Öjebyn are built on unfree land, meaning that the owners do not own the land on which they are situated. The Piteå parish primarily owns the land on which the church cottages are situated within the Church Town, while individual cottage owners hold ownership rights to their respective units [47].

Additionally, the cottages commonly used have multiple co-owners, who collectively share the responsibility for maintaining the exterior structure of the building [51,54].

2.3. Typological and Spatial Analysis

A critical aspect of understanding the preconditions for integrating PV involves analysing the architectural typology to determine which building elements can accommodate discrete, reversible, or fully integrated photovoltaic (PV) solutions. Architectural characteristics such as roof shapes, materials, and facade treatments define not only the technical feasibility but also the visual acceptability of PV installations, given the imperative to preserve historical character [9,22].

Although the size of the church cottages varies, they are connected by their low height, gable roofs, and simple designs. Green infrastructure plays a significant role in the landscape from both cultural and historical perspectives. It emphasises the function of church cottages as dwellings, as they are tightly built near each other, and distinguishes them from former adjacent stables or other built structures, such as the Church and newer buildings surrounding the church town's central core (Figure 1).

The mixed typology of church cottages contributes to a distinctive, low-key skyline that defines the place's identity. Many church cottages are multi-chambered, with two floors, and are often mixed with one-story and two-chamber cottages. The result is a variegated roof landscape with varying roof ridge directions and variations in house heights and widths.



Figure 1. The image shows church cottages with contemporary adaptations, such as new metal roofs and previous, rusty ones. It also features a zone of green fields marking the boundary to the rest of the Öjeby village. Source: The authors.

The cottages exhibit adaptations to contemporary requirements, featuring modern additions such as downpipes, gutters, and metal-sheeted roofs. Some measures are acceptable, others are not [55]. Building permits are needed to integrate PV on buildings in the protected area, and consequently, PVs are not allowed in the Öjeby heritage district. However, this policy is not always adopted or followed (Figure 2).



Figure 2. Photovoltaics on the shutters of a church cottage in Öjeby, a church town in Piteå, show some interest in integration for small-scale energy production. Source: The authors.

2.4. PV Technology in Geographical and Architectural Context

Local climatic conditions impose additional constraints and opportunities that shape the preconditions for integration. Northern Sweden experiences low winter sun angles and substantial snow loads yet benefits from high summer solar yields and enhanced PV efficiency in cold climates. Consequently, PV systems must be adapted to local solar geometry, snow behaviour, and visual integration requirements to ensure both technical performance and cultural compatibility [56].

The northern region of Sweden has unique climate conditions that may limit the feasibility and acceptance of PV integration. The area benefits, however, from favourable solar values, offering opportunities for solar energy production [57]. Although the region

experiences limited sunlight during the dark winters, the rich amount of sunlight during the spring and summer months may balance out the electricity production.

One potential approach is to install photovoltaic (PV) panels vertically on facades and outdoor solar installations. Vertical Bifacial PV (VBPV) systems offer alternative placement options, particularly for avoiding conflicts with protected surfaces. VBPV systems are argued to be more efficient than traditionally mounted systems [55]. This approach also helps prevent snow accumulation, a common issue with roof-mounted panels. Vertically mounted panels can capture reflections from the snow and benefit from the sun's low angle during winter. Additionally, polycrystalline PV panels have demonstrated good efficiency in cold climates [58,59].

Another approach to integrating PV systems in heritage districts may be to avoid applying them directly to the surfaces of buildings, such as roofs, exterior walls, and shutters. Instead, PV systems can be used as standalone, small-scale installations, such as advertising signs, lampposts, and similar structures located between buildings. Initiatives have also been undertaken to implement artistic and reversible photovoltaic solutions, such as self-propelled advertising columns and "sun flags" near or on monumental buildings [60].

2.5. The Stakeholders

Built artefacts can be considered stakeholders, as anyone who influences or is affected by decisions may be subject to ethical considerations [61]. Therefore, the primary stakeholder is the Church Town itself, comprising the heritage district, the church cottages, and Öjeby Church.

In Piteå, the Cultural Environment Program outlines the municipality's overarching cultural environment strategy, which includes programs that detail preservation measures for built environments with exceptional heritage values [62]. Guidelines for preservation are reflected in the in-depth description of national interest (in the context of cultural heritage) for Öjeby Church Town. This document details the designation of Öjeby Church Town as a nationally significant cultural environment, including its inclusion in both municipal and regional cultural environment plans [63].

The most important stakeholders come together in the Church Town Council, which serves as a forum for discussion focused on preserving and developing the district. The council addresses a variety of issues related to past, present, and future initiatives for the church town.

One stakeholder representative involved in the council is the Church Town Cottage Owner Association, which coordinates resources and aims to address and resolve various issues from the residents' perspective.

The Piteå municipality is a key stakeholder, as it is the official authority for Öjebyn and the Church Town, both of which fall within its jurisdiction. In Sweden, local authorities have the exclusive right to determine how land can be used and developed [64]. This responsibility rests with municipal officials, including city architects, planning administrators, and building permit officers.

The County Museums are among the most relevant stakeholders in preserving heritage areas in Sweden. These museums collaborate closely with county administrative boards, local authorities, and the general public. They play a central role in preserving, using, and developing the cultural environment [65]. The County Museum of Norbotten (Norrbottens Museum) oversees the county's heritage.

The Piteå Museum Association conducts its activities on a local level, with partial funding from the municipality supporting its operations. For example, the museum participates in consultation processes and expert groups, such as those involved in developing detailed plans, in collaboration with the municipality [66].

The Church of Sweden, through Piteå parish, is responsible for managing, preserving, and conducting activities in the area and serves as a key player in the Church City Council, often acting as the project owner for larger investments in protecting the church town [67].

The Solander Society is a non-profit association that aims to preserve and develop the cultural environment in Öjeby Church Town [68]. They are often involved in events such as the Church Town Day.

It is argued that practice-based approaches have the potential to articulate knowledge and politics actively through design, as they can provide essential knowledge and other forms of thinking that are lacking in other disciplines [69]. In our study, community members in the heritage district play a crucial role as key stakeholders, as their acceptance is essential for the heritage significance; consequently, a design workshop will be created for them.

However, individuals cannot be classified as stakeholders unless they possess ownership and comply with municipal conservation guidelines. Today, being a stakeholder means having some level of ownership of a cottage [70]. Therefore, residents of Öjeby Church Town, including those who temporarily reside in cottages, comprise the group of qualified community stakeholders also, as a living church town, a tradition exists today where many owners identify themselves as stewards of the cultural environment, driven by both collective norms and personal motives, including heritage, homeliness, and recreation [51].

3. Materials and Methods

A tailored participatory approach for heritage districts may be crucial for examining the perspectives and acceptance of various stakeholders, including residents' needs and motivations. The method aims to address the research question of how the perspectives of residents as prosumers can be effectively integrated into decision-making processes for small-scale photovoltaic (PV) integration in heritage districts.

Our study approach consists of several steps (Figure 3).

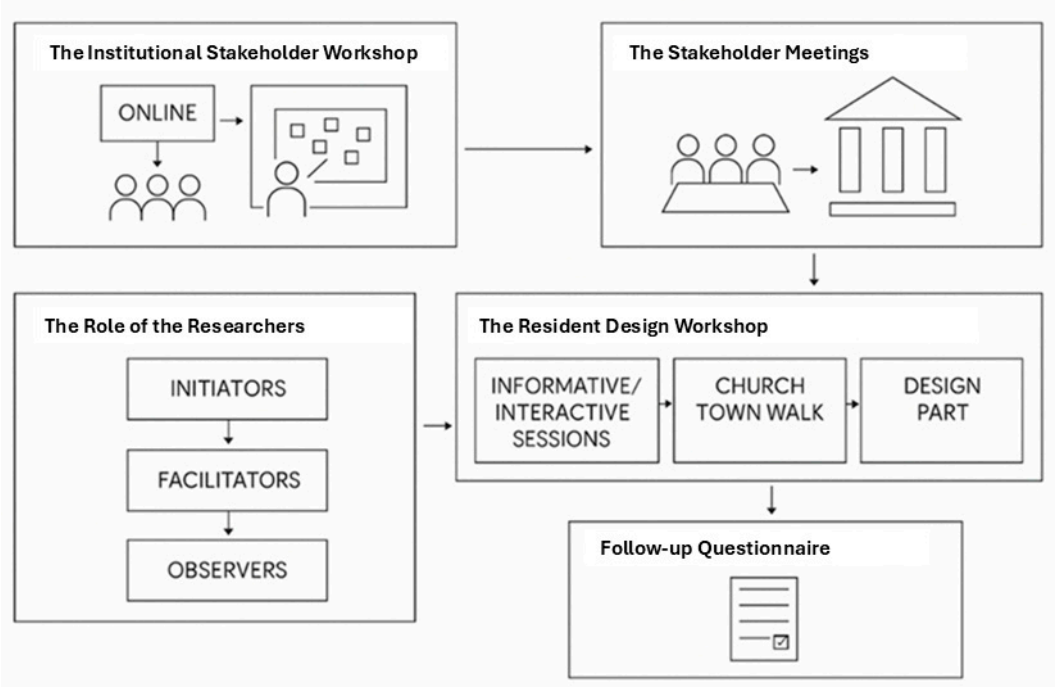


Figure 3. The flow chart displays the tailored participatory method employed for this study.

3.1. The Institutional Stakeholder Workshop

An important initial step in this study was to organise an online workshop. This approach was chosen because it effectively addresses contextual limitations and facilitates participatory input. It is argued that digital collaboration tools provide greater flexibility and support asynchronous contributions, which are especially beneficial during health or logistical crises [71]. Invited stakeholders included experts from the local authority and the North Energy Agency. This workshop served as a platform for collaborative discussions, idea generation, and the establishment of foundational concepts, primarily involving representatives from stakeholder groups.

The digital mural software facilitated visual collaboration and allowed for saving the workshop content in a PDF format, ensuring outcomes were preserved for future reference. The use of digital tools, such as Mural software (version 6.7.2), enhances inclusivity, accessibility, and documentation of workshop processes, which are essential for transparency and iterative analysis in co-creative research [41,72].

3.2. The Stakeholder Meetings

To interact with stakeholders, we physically attended meetings at Church Town Council and the Solander Association as observers. It follows that attending stakeholder meetings as observers is a form of non-intrusive engagement that enables researchers to gather contextual insights and understand social dynamics within real-world decision-making environments [73]. Additionally, expert interviews and informal meetings with local municipal and museum staff provided localised, tacit, and historical knowledge critical in heritage-based energy planning.

3.3. The Resident Design Workshop

This study places significant emphasis on participatory design by actively engaging residents to explore their viewpoints and levels of acceptance regarding the design process. Therefore, it is essential to employ empowerment-focused participation, enabling users to control the design process by defining both problems and solutions [74].

To achieve this, conducting a customised participatory design workshop tailored to the community's unique needs and circumstances is essential, as this approach aims to ensure residents' voices are heard and integrated into the design outcomes.

The resident design workshop was planned to be divided into three main sessions. First, the informative and Interactive sessions, secondly, the church town walk, and finally, the design activity (Table 1).

Table 1. The table outlines the three main components of the tailored participatory design approach, along with the associated activities and objectives.

Workshop Component	Activity	Objective
1. The Informative & Interactive Session	Presentations on heritage values, energy communities, and PV technologies	Introduce key concepts and gather initial perceptions
2. The Church Town Walk	Site visits to three selected locations with historic/modern intersections	Prompt ideas for spatial integration of PV through contextual learning
3. The Design Part	Group model-making using analogue cardboard models and cut-out panels	Generate spatial PV placement ideas; assess acceptance and visibility impact

Researchers, in collaboration with various participating stakeholders, should consider both explicit and tacit knowledge and utilise it in transdisciplinary collaboration and communication. Michael Polanyi [75] famously stated, "We can know more than we can tell", emphasising that tacit knowledge underlies all explicit knowledge. It is elaborated that tacit knowledge is personal and context-specific, often acquired through personal experience and practice rather than formal instruction.

Various modes of participation ensure that stakeholders, particularly end-users, are actively involved in the design process. Therefore, our role involves engaging participating community members by acting as (1) Initiators, (2) Facilitators and (3) Observers. Facilitators play a crucial role in guiding collaboration and translating user input into actionable insights [76]. Meanwhile, initiators help frame the problem and align the design with stakeholder needs, and observers collect data on user behaviours and preferences [77].

The following step was to recruit participants from the Öjebyn community. For this reason, announcements were made on social media groups related to Öjebyn, and flyers for the event were posted on public notice boards.

Workshop Component 1: The Informative and Interactive Session.

These sessions were designed with a pedagogical aim to equip participants with the foundational knowledge necessary for meaningful participation in the subsequent design activities. These sessions emphasised active learning through dialogue and reflection [78], allowing participants to engage critically with the complexities of integrating PV technologies into Öjebyn's heritage environment. The sessions also align with community empowerment and the co-construction of knowledge [79]. By framing discussions around questions such as "Is it possible to combine cultural heritage and new photovoltaic technologies?" participants were encouraged to articulate their needs, values, and visions regarding renewable energy transitions.

This pedagogical framework is essential in participatory design, as informed participants are more likely to contribute constructively and confidently to co-creative processes. The use of interactive presentations, open dialogue, and collaborative tools aligns with educational approaches that emphasise situated learning, where knowledge is acquired in the context in which it is applied [80].

Workshop Component 2: The Church Town Walk.

The walk was employed for experiential learning and collective reflection. Such methods are increasingly recognised in urban pedagogy and participatory planning for their ability to contextualise abstract concepts within the lived and physical landscape [81,82]. The walk facilitated immersion in the heritage environment, allowing participants to observe the architectural characteristics and spatial constraints of Öjebyn firsthand. This method also supports place-based learning, where participants generate insights through embodied engagement with their surroundings [83].

Before the walk, participants were presented with a speculative future scenario: "By 2040, Sweden will generate all its energy from renewable sources... Your task includes creating solar installations that contribute energy to the grid." Scenarios like this are a recognised strategy in participatory futures and sustainability science, as they enable communities to explore and debate multiple possible futures, assess trade-offs, and envision preferred pathways [84]. Scenario-based thinking also enhances deliberation and aligns with anticipatory governance, supporting long-term planning in culturally sensitive contexts [44].

Workshop Component 3: The Design Part.

The design activity involved visual assessment tasks in which participants evaluated various PV integration scenarios, including church cottages, green structures, and grey infrastructure, using visual tools. This approach has a significant impact on public acceptance of solar technologies, particularly in heritage areas [85]. The inclusion of building-applied

photovoltaics (BAPV) was intentional to reflect feasible and minimally invasive integration strategies. Visual simulations enabled comparison across design variants and facilitated nuanced discussions regarding visibility, cultural compatibility, and energy potential.

This method aligns with the principles of value-sensitive design, which integrates user values, visual preferences, and ethical considerations into technical decision-making [86]. It also addresses the need for transdisciplinary co-creation in energy heritage planning, where community acceptance and cultural continuity are crucial to the success of implementation.

3.4. Follow-Up Questionnaire

An online questionnaire was created to get (1) background information of the participants, (2) their needs and ambitions regarding photovoltaic integration, (3) the residents' value of photovoltaic integration in Öjeby Church Town, and (4) Residents' value of Öjeby Church Town, e.g., how should the cultural heritage be managed in the future? The questionnaire was designed to be answered before and after the design event to analyse how the participants' views changed as a result of the workshop.

4. Results

4.1. Findings of the Institutional Stakeholder Workshop

Photovoltaics are not yet a local priority. While Piteå has a strong commitment to renewable energy in general, photovoltaics (PV) does not currently appear to be a strategic focus at the local level. This suggests that other forms of renewable energy may be prioritised, or that there is still some hesitation or uncertainty around the role of PV in the region's energy future.

There is a disconnect between innovation and local strategy. A notable gap exists between the development of innovative PV technologies at the national and international levels and their consideration in local energy strategies. This may indicate a need for improved communication or coordination between different levels of policy and practice.

4.2. Findings of the Stakeholders Meetings

National policies provide structure but constrain local innovation and adaptation, including the integration of PV systems.

One of the most striking paradoxes uncovered during site visits and stakeholder meetings is the approach to roofing materials. Historically, traditional wooden roofs were replaced with metal sheet roofs to comply with fire regulations and prevent the spread of fire. However, these metal roofs, now aged and rusty, are considered integral to the visual and material identity of the church cottages and are not recommended for replacement.

This shows a dilemma where restoring the "original" look of the heritage district would conflict with practical considerations of maintenance and safety. At the same time, it demonstrates how contemporary materials or elements, introduced to replace the original ones, can become part of a new identity for a heritage site and, therefore, how the concept of authenticity can evolve.

Some issues related to our study were discussed at the Church Town Council meetings, including: Signage projects and inadequately renovated church cottages.

Currently, the council are investigating the optimal locations for sign placement to ensure consistency throughout the area. There are growing problems with cottages that were inadequately renovated in the 1970s.

4.3. Findings of the Resident Design Workshop

The resident workshop on integrating solar panels into Öjebyn's culturally and historically valuable environments revealed several key insights and challenges. It assessed

how spatial layout and design elements influence the acceptance and effectiveness of photovoltaic integration.

Findings on the Informative and Interactive Session.

The discussion revealed opinions on what is acceptable and the feasibility of integration actions. For example, window shutters have a specific "language" and high historical value. Carelessly applying solar cells to them can cause them to lose their unique expression and original function (Figure 2). If they are to be installed, they should be integrated with colour matching. Nonetheless, some residents had objections to this, as it can be costly, and the electricity produced may not justify the investment.

Findings on the Church Town Walk.

The walk was organised with the primary aim of inspiring residents and informing them about the possibilities for the upcoming design phase. This allowed residents to experience the site collectively, foster dialogue, and develop a shared understanding of the site's spatial qualities and constraints. Walking through the environment helped them to engage with tangible heritage values and visualise potential interventions in situ.

One of the key criteria for integration concerned the use of building-applied photovoltaics (BAPVs), which should employ a visibility-sensitive approach across various scenarios. The objective was to identify residents' subjective preferences regarding the placement of solar panels on the roofs of the church cottages while preserving critical sightlines that define the heritage experience of the church town. Additionally, the walk sought to determine the extent to which PV could be integrated without compromising these visual qualities.

Findings on the Design Part.

The workshop presented different scenarios with varying numbers of BAPV panels to illustrate the relationship between electricity production and consumption more effectively. During the design phase, participants worked in groups using analogue sketch models representing the site's plots and houses. On these models, the task involved attaching coloured cardboard cut-outs to indicate proposed PV placements (Figure 4a). This approach encouraged creative exploration of strategies for integrating PV technologies without altering building envelopes and while respecting critical sightlines.



(**a**) (**b**)

Figure 4. (**a**) shows one (of three) models the participants worked with. (**b**) shows a newly mounted metal-roofed church cottage. Source: The authors.

Ideas included small-scale PV installations on green infrastructure and rotating solar sculptures that can track the sun to increase efficiency. These installations were also positioned on surrounding buildings rather than on the church cottages themselves. This strategy aimed to minimise the visual and structural impact on heritage buildings while simultaneously enhancing the functionality and aesthetic quality of otherwise underutilised

spaces. As such, it was regarded as a viable alternative to the direct application of PV on historically significant structures (Table 2).

Table 2. Summary of design criteria and the residents' preferences.

Design Criterion	Resident Preference
Visual Integration	PV should be as inconspicuous as possible (colour-matched, hidden)
Cultural Respect	Historical elements must retain their symbolic function (e.g., shutters)
Location Strategy	Focus on non-core heritage buildings and public/shared infrastructure
Technology Flexibility	Acceptance of BAPV and creative alternatives over BIPV when cost-effective

4.4. Findings of the Follow-Up Questionnaire

The questionnaire revealed that participants were generally restrictive in their approach to PV integration. However, a subsequent questionnaire indicated a more positive attitude among the participants. However, this result could not be validated due to statistical limitations, as it involved a small number of participants and different individuals answering before and after the intervention.

5. Discussion

This study aimed to explore how the perspectives of residents, as potential prosumers, can be effectively integrated into decision-making processes for small-scale PV integration in heritage districts, using Öjeby Church Town as a case study.

Acknowledging historical layers and cultural practices is critical for contemporary design and planning. Interventions that reflect and respect these traditions contribute to cultural sustainability, reinforcing local identity and ensuring that heritage environments remain socially meaningful [46,49]. Conversely, designs that ignore these embedded values risk alienating communities and undermining the authenticity that gives these spaces their enduring significance [32,38].

As a result, this case study extends beyond a purely local inquiry; its findings may inform broader discussions on heritage policy, energy transition, and participatory urbanism [7,8,34]. By examining how a heritage-related settlement navigates PV integration, the case provides insights that can guide similar heritage sites in achieving sustainability without compromising cultural values.

5.1. The Institutional Stakeholder Workshop

Preserving Öjeby's heritage district presents a complex interplay between historical conservation, modern regulations, and evolving community needs [22,29]. However, the municipality's current renewable energy strategy does not explicitly prioritise PV integration, which may reflect broader strategic considerations or resource allocation preferences [1,3].

Furthermore, the district's development plan lacks specific protection provisions, which has led to a paradox where the municipality recommends against installing photovoltaic (PV) systems, even on modern houses within the core of the heritage area [28]. This stance highlights the tension between maintaining historical aesthetics and embracing sustainable energy solutions [9,21].

These observations highlight a valuable area for future research: exploring the factors that may be limiting the uptake of PV in municipalities that are already strong in

renewable energy production [30,31]. Possible barriers may include economic considerations, regulatory frameworks, or community perceptions regarding the compatibility of aesthetics and heritage.

5.2. The Stakeholders Meetings

The stakeholders' meetings reveal significant complexities. A key tension emerges between preserving the district's historical authenticity and adopting PV technologies [32].

The fact that modern metal sheet roofs are permitted (Figure 4b) while the installation of integrated photovoltaic systems (BIPV) is prohibited highlights further controversial, if not contradictory, outcomes [27,29]. While modern metal roofs are seen as adaptive and coherent with the cottages' aesthetic, BIPVs are viewed as intrusive despite their potential to provide sustainable energy [12]. This preference highlights a broader challenge in heritage conservation: striking a balance between preserving historical character and adopting modern PV technologies (Table 3).

Table 3. The results from the case study analysis, stakeholder workhops, and meetings display some of the challenges in integrating PV in Öjeby Church Town.

Challenge	Observation	Implication
PV Restrictions	PV systems discouraged, even on non-historical buildings in heritage core	Highlights regulatory hesitance towards integrating new technologies
Metal Roofing	Metal roofs replaced wooden ones due to fire safety; now considered part of the historical look	Shows how newer materials can gain cultural acceptance over time
BIPV Controversy	Building-integrated photovoltaics (BIPV) is forbidden, while metal roofs are allowed	Reveals contradictory policies toward modern materials

This paradox highlights how heritage authenticity is not fixed but can evolve to incorporate modern interventions that become culturally embedded over time [44,53].

Statutory emphasis on preserving the historical and cultural identity of the district often limits opportunities for integrating contemporary sustainable technologies, such as PV systems, that could contribute to local energy transitions [28,29]. These policy constraints suggest a need for more nuanced regulatory frameworks that balance the dual imperatives of heritage preservation and environmental sustainability, potentially through adaptive guidelines allowing context-sensitive PV applications [24,25].

Moreover, this points to an essential pathway for incremental integration of renewable energy technologies within heritage districts by embedding PV within ancillary structures or functional elements (e.g., signage), thus minimising visual impact while adding value [60].

Consequently, both issues could be addressed through small-scale PV solutions if they were aesthetically and technically feasible, manageable, and cost-effective.

5.3. The Resident Design Workshop

The resident design workshop confirmed that participatory processes are essential to balancing sustainability and heritage conservation [33,35,76]. Community-led proposals, if guided and informed, can lead to innovative yet respectful PV integration strategies [36,86].

Participants recognised the need to preserve aesthetic values and proposed thoughtful, non-invasive PV solutions. There was strong interest in placing PV on green/grey infras-

tructure over heritage buildings, and conditional acceptance of PV on cottages, provided it was well-hidden and justified by visibility analysis [20,60].

Overall, the participatory method increased awareness, acceptance, and a sense of shared responsibility [74].

The design activity encouraged creative problem-solving and reinforced residents' sense of agency, resulting in community-led proposals [33]. Such empowerment has been shown to enhance local motivation and reduce resistance to new technologies in energy transitions [26,31].

5.4. A Proposed Transdisciplinary Participatory Approach

To support the urban energy transition, building regulations, conservation policies, and energy tax legislation can be strategically adapted to enable the sensitive integration of photovoltaic (PV) systems within heritage districts [24,25].

In the case of Öjebyn's Church Town, one should also navigate the complex governance and ownership structures of the church cottages, as many are subject to co-ownership arrangements that require consensus for modifications and can complicate the adoption of new technologies, such as PV systems [48,52].

Any PV integration strategy must also respect these historical and cultural values while enabling the transition toward renewable energy [7,50]. A central coordination platform for these diverse interests may be the Church Town Council [62]. Integrating PV adoption into the council's agenda could provide a practical governance pathway for balancing heritage conservation with energy transition goals.

PV integration strategies in the northern Swedish context must also consider the region's specific geographical and architectural conditions [55,56,58,59]. Selecting among these technological options requires collective deliberation within established governance frameworks, ensuring that technical feasibility, heritage preservation, and community values are all addressed in a coordinated decision-making process [34].

Achieving this goal may require developing a transdisciplinary approach that involves not only experts from the fields of energy and architecture but also the active participation of residents as future end-users [40,74].

Participatory Design can be seen as a mediator, as it bridges stakeholder tensions and surfaces hidden knowledge (both tacit and local) [33,35]. Furthermore, it can align technological, cultural, and architectural dimensions, ensuring that final decisions reflect both expert and community input.

There is a continued need to advance participatory design practices with residents in Urban Living Labs (ULLs) [44,45]. Future research could focus on developing a virtual testbed, further refining the method through an iterative, locally anchored process involving a diverse range of stakeholders.

This collaborative and inclusive approach, grounded in the cultural and historical significance of church towns, supported by governance coordination, and informed by context-specific PV technologies, is expected to generate synergies that enhance the built environment while respecting the unique heritage value of these districts.

A transdisciplinary initiative should involve a variety of stakeholders in an Urban (energy) Living Lab environment (Figure 5). This U(e)LL should include:

1. Residents in the heritage district (D)
2. Local authorities (M), including urban planners and building permit officials.
3. Experts in heritage conservation (E), such as representatives from local and regional museums and conservationists of the built environment.

Other experts involved could include:

4. PV developers who are responsible for the technical implementation of solar energy systems adapted to the climatic and architectural conditions of the district.
5. Researchers with expertise in facilitating design processes.
6. Architects who can contribute to spatial and typological analysis.

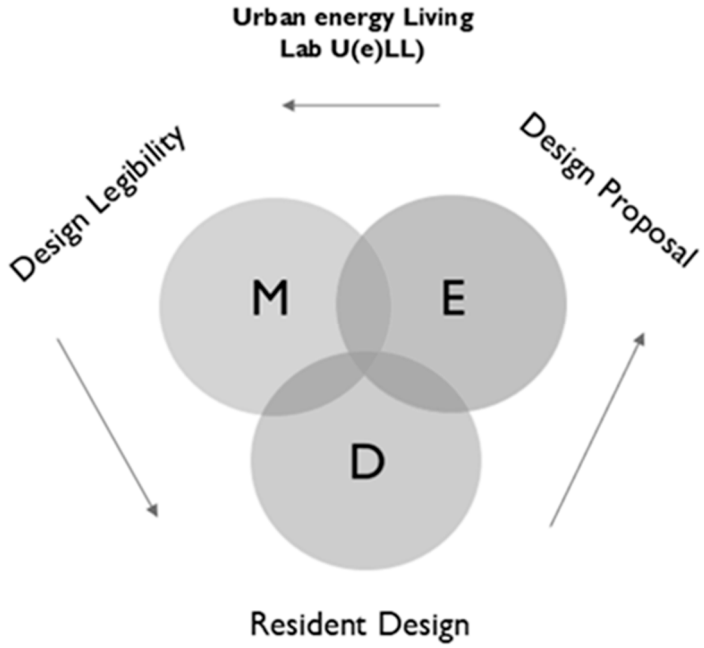


Figure 5. The diagram displays a proposed transdisciplinary approach for value-sensitive integration of small-scale PV.

Resident design emphasises ownership of the process at the earliest stage, ensuring that design interventions respond to community needs, aspirations, and socio-cultural identities [41]. The design proposal represents an intermediary product that bridges residents' visions and the practical requirements of implementation, while balancing innovation with contextual appropriateness [42]. Design legibility entails visual and narrative coherence, demonstrating how interventions respect cultural landscapes while meeting contemporary energy demands [43].

The collaborative framework proposed, involving local authorities, experts in heritage conservation, PV developers, researchers, architects, and residents, embodies a holistic approach that transcends traditional sectoral silos and addresses complex socio-technical challenges.

6. Conclusions

This study examined the integration of photovoltaic (PV) technologies within the heritage context of a sensitive heritage site, addressing how the perspectives of residents, as potential prosumers, can be effectively incorporated into decision-making processes.

The findings reveal that tensions persist between preserving visual and cultural integrity and adopting renewable energy solutions, reflecting a global challenge in protected heritage districts. While technological advances in PV enable context-sensitive applications, uptake is hindered by restrictive regulatory frameworks, complex governance structures, and strong community preferences for maintaining historical aesthetics.

The participatory design approach applied here proved essential for integrating residents' perspectives. By engaging residents in co-creative workshops and visibility-based design discussions, the process enabled them to articulate aesthetic priorities, propose spatially flexible PV solutions, and evaluate trade-offs between heritage values and energy

benefits. This participatory governance model enhanced residents' sense of agency fostered mutual understanding between stakeholders and facilitated the co-production of solutions that balance authenticity with sustainability.

The resulting collaborative framework, linking local authorities, heritage experts, PV developers, researchers, and residents, offers a replicable pathway for integrating prosumer perspectives into small-scale PV planning in heritage settings. Effective integration requires adaptive regulatory guidelines that accommodate context-sensitive PV designs, transparent decision-making processes that include visibility and impact assessments, and governance structures capable of mediating diverse interests. Ultimately, embedding resident participation within formal heritage energy planning processes ensures that renewable energy adoption supports both the preservation of cultural identity and the transition toward a sustainable energy future.

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Institutional Review Board Statement: Ethical review and approval were waived for this study because it involved minimal risk to participants and met the criteria for exemption under institutional ethical guidelines. The study consisted of an anonymous online questionnaire administered before and after a workshop. Participation was voluntary, and no sensitive personal data was collected; responses were recorded without any identifying information. All information and data are processed in accordance with the GDPR. Only six participants took part in the study, and all were informed about the purpose of the research and their right to withdraw at any time.

Data Availability Statement: The dataset is available on request from the authors.

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